SGT UNIVERSITY

School of Engineering and Technology

Department of CSE

DATABASE MANAGEMENT SYSTEMS LABORATORY

Experiment 2

CREATE TABLE and ALTER TABLE (Rename Column)

Field	Details
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Programme	B.Tech CSE (AI/ML)
Section	Section - C
Lab Date	26 August 2026
Dataset	Employees — 6 sample rows × 4 attributes
Methods	CREATE TABLE, ALTER TABLE ... RENAME COLUMN

1. Aim

To create an `employees` table with attributes `empID`, `empName`, `department` and `salary`, insert sample records, and then rename the column `department` to `faculty` using `ALTER TABLE`.

2. Theory

2.1 CREATE TABLE

A relational table is defined by a name, a list of attributes, and constraints that protect the data. `PRIMARY KEY` makes `empID` unique and not null. `NOT NULL` rejects missing names and faculty values. `CHECK (salary > 0)` rejects non-positive salaries. `NUMERIC(10, 2)` stores pay as a decimal with two fractional digits.

2.2 ALTER TABLE RENAME COLUMN

In SQLite and MySQL 8+, `ALTER TABLE ... RENAME COLUMN` changes a column's name without rewriting the stored values. After `RENAME COLUMN department TO faculty`, every row still holds the same school name; only the heading of that column becomes `faculty`. This is a schema change, not an `UPDATE`. Microsoft SQL Server does not accept that syntax; the equivalent is `EXEC sp_rename 'employees.department', 'faculty', 'COLUMN'` (see `employees.sqlserver.sql`).

3. Schema

The table is created with four attributes. After the rename, the third attribute is `faculty` instead of `department`.

Attribute	Type	Constraint	Role
empID	INTEGER	PRIMARY KEY	Employee identifier
empName	TEXT	NOT NULL	Employee name
department → faculty	TEXT	NOT NULL	School / faculty name
salary	NUMERIC(10,2)	NOT NULL, CHECK > 0	Monthly salary

Sample faculty values include **School of Engineering and Technology** (the faculty named on this report's cover). The cover also lists **Department of CSE** as the student's home department; that heading is letterhead, not a fifth column in the table.

4. Procedure

1. Drop `employees` if it already exists, so the script can be re-run.
2. Create the table with `empID`, `empName`, `department` and `salary`.
3. Insert six sample employees.
4. Display all rows and the column list with `PRAGMA table_info`.
5. Rename `department` to `faculty`.
6. Display all rows and the column list again to confirm that only the name changed.

5. Source Code

```
-- ============================================================
-- DBMS Lab · 26-08-2026
-- Employees table: CREATE, seed, then RENAME department → faculty
```

```sql
-- SQLite  (sql.js compiler / DB Browser / sqlite3)
-- For the class SQL Server, run employees.sqlserver.sql instead.
-- ============================================================

DROP TABLE IF EXISTS employees;

-- ------------------------------------------------------------
-- 1. Create the employees table
-- ------------------------------------------------------------
CREATE TABLE employees (
    empID      INTEGER PRIMARY KEY,
    empName    TEXT NOT NULL,
    department TEXT NOT NULL,
    salary     NUMERIC(10, 2) NOT NULL CHECK (salary > 0)
);

-- ------------------------------------------------------------
-- 2. Insert sample rows (department holds the faculty/school name)
-- ------------------------------------------------------------
INSERT INTO employees (empID, empName, department, salary) VALUES
    (1, 'Ananya Sharma',  'School of Engineering and Technology', 92000.00),
    (2, 'Rohan Mehta',    'School of Management',                 78000.00),
    (3, 'Priya Nair',     'School of Law',                        85000.00),
    (4, 'Vikram Singh',   'School of Agricultural Sciences',      76000.00),
    (5, 'Fatima Khan',    'School of Engineering and Technology', 88000.00),
    (6, 'Arjun Patel',    'School of Medical Sciences',           95000.00);

-- ------------------------------------------------------------
-- 3. Verify the table BEFORE the rename
-- ------------------------------------------------------------
SELECT * FROM employees;
PRAGMA table_info(employees);

-- ------------------------------------------------------------
-- 4. Rename the column department → faculty
--    Row values are unchanged; only the column name changes.
-- ------------------------------------------------------------
ALTER TABLE employees RENAME COLUMN department TO faculty;

-- ------------------------------------------------------------
-- 5. Verify the table AFTER the rename
-- ------------------------------------------------------------
SELECT * FROM employees;
PRAGMA table_info(employees);
```

6. Output

The script was executed in SQLite. `PRAGMA table_info` columns are `cid` (column index), `name`, `type`, `notnull`, `dflt_value` and `pk`.

```
SQL> DROP TABLE IF EXISTS employees;
-- OK

SQL> CREATE TABLE employees (
         empID      INTEGER PRIMARY KEY,
         empName    TEXT NOT NULL,
         department TEXT NOT NULL,
         salary     NUMERIC(10, 2) NOT NULL CHECK (salary > 0)
     );
-- OK

SQL> INSERT INTO employees (empID, empName, department, salary) VALUES
         (1, 'Ananya Sharma',  'School of Engineering and Technology', 92000.00),
         (2, 'Rohan Mehta',    'School of Management',                 78000.00),
         (3, 'Priya Nair',     'School of Law',                        85000.00),
         (4, 'Vikram Singh',   'School of Agricultural Sciences',      76000.00),
         (5, 'Fatima Khan',    'School of Engineering and Technology', 88000.00),
         (6, 'Arjun Patel',    'School of Medical Sciences',           95000.00);
-- 6 row(s) inserted

SQL> SELECT * FROM employees;
empID  empName        department                            salary
-----  -------------  ------------------------------------  ------
1      Ananya Sharma  School of Engineering and Technology  92000
2      Rohan Mehta    School of Management                  78000
3      Priya Nair     School of Law                         85000
4      Vikram Singh   School of Agricultural Sciences       76000
5      Fatima Khan    School of Engineering and Technology  88000
6      Arjun Patel    School of Medical Sciences            95000

SQL> PRAGMA table_info(employees);
cid  name        type            notnull  dflt_value  pk
---  ----------  --------------  -------  ----------  --
0    empID       INTEGER         0                    1
```

```
1    empName     TEXT            1                    0
2    department  TEXT            1                    0
3    salary      NUMERIC(10, 2)  1                    0

SQL> ALTER TABLE employees RENAME COLUMN department TO faculty;
-- OK

SQL> SELECT * FROM employees;
empID  empName       faculty                             salary
-----  ------------  ----------------------------------  ------
1      Ananya Sharma  School of Engineering and Technology  92000
2      Rohan Mehta    School of Management                  78000
3      Priya Nair     School of Law                         85000
4      Vikram Singh   School of Agricultural Sciences       76000
5      Fatima Khan    School of Engineering and Technology  88000
6      Arjun Patel    School of Medical Sciences            95000

SQL> PRAGMA table_info(employees);
cid  name     type            notnull  dflt_value  pk
---  -------  --------------  -------  ----------  --
0    empID    INTEGER         0                    1
1    empName  TEXT            1                    0
2    faculty  TEXT            1                    0
3    salary   NUMERIC(10, 2)  1                    0
```

7. Results

Stage	Third column	Row count	Values changed?
Before ALTER	department	6	No — original schema
After ALTER	faculty	6	No — names only

All six employees remain. School names such as School of Engineering and Technology are unchanged. The schema listing is the proof: `cid = 2` is named `department` before the statement and `faculty` afterwards.

8. Conclusion

An employees relation can be created with a primary key, not-null attributes and a salary check, then adjusted with `ALTER TABLE ... RENAME COLUMN`. Renaming `department` to `faculty` updates the schema only. The stored values do not change, which is the correct way to relabel an attribute without rewriting the table.